Supervised vs. Unsupervised Learning: What's the Difference?

Artificial intelligence

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In this article, we'll explore the basics of two data science approaches: supervised and unsupervised. Find out which approach is right for your situation.

The world is getting "smarter" every day, and to keep up with consumer expectations, companies are increasingly using machine learning algorithms to make things easier. You can see them in use in end-user devices (through face recognition for unlocking smartphones) or for detecting credit card fraud (like triggering alerts for unusual purchases).

Within artificial intelligence (AI) and machine learning, there are two basic approaches: supervised learning and unsupervised learning. The main difference is one uses labeled data to help predict outcomes, while the other does not. However, there are some nuances between the two approaches, and key areas in which one outperforms the other. This post will clarify the differences so you can choose the best approach for your situation.

What is supervised learning?

Supervised learning is a machine learning approach that's defined by its use of labeled datasets. These datasets are designed to train or "supervise" algorithms into classifying data or predicting outcomes accurately. Using labeled inputs and outputs, the model can measure its accuracy and learn over time.

Supervised learning can be separated into two types of problems when data mining: classification and regression:

- **Classification** problems use an algorithm to accurately assign test data into specific categories, such as separating apples from oranges. Or, in the real world, supervised learning algorithms can be used to classify spam in a separate folder from your inbox. Linear classifiers, support vector machines, decision trees and random forest are all common types of classification algorithms.
- **Regression** is another type of supervised learning method that uses an algorithm to understand the relationship between dependent and independent variables. Regression models are helpful for predicting numerical values based on different data points, such as sales revenue projections for a given business. Some popular regression algorithms include linear regression, logistic regression and polynomial regression.

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What is unsupervised learning?

Unsupervised learning models are used for three main tasks: clustering, association and dimensionality reduction:

- **Clustering** is a data mining technique for grouping unlabeled data based on their similarities or differences. For example, K-means clustering algorithms assign similar data points into groups, where the K value represents the size of the grouping and granularity. This technique is helpful for market segmentation, image compression, etc.
- **Association** is another type of unsupervised learning method that uses different rules to find relationships between variables in a given dataset. These methods are frequently used for market basket analysis and recommendation engines, along the lines of "Customers Who Bought This Item Also Bought" recommendations.
- **Dimensionality reduction** is a learning technique used when the number of features (or dimensions) in a given dataset is too high. It reduces the number of data inputs to a manageable size while also preserving the data integrity. Often, this technique is used in the preprocessing data stage, such as when autoencoders remove noise from visual data to improve picture quality.

The main difference between supervised and unsupervised learning: Labeled data

The main distinction between the two approaches is the use of labeled datasets. To put it simply, supervised learning uses labeled input and output data, while an unsupervised learning algorithm does not.

In supervised learning, the algorithm "learns" from the training dataset by iteratively making predictions on the data and adjusting for the correct answer. While supervised learning models tend to be more accurate than unsupervised learning models, they require upfront human intervention to label the data appropriately. For example, a supervised learning model can predict how long your commute will be based on the time of day, weather conditions and so on. But first, you'll have to train it to know that rainy weather extends the driving time.

Unsupervised learning models, in contrast, work on their own to discover the inherent structure of unlabeled data. Note that they still require some human intervention for validating output variables. For example, an unsupervised learning model can identify that online shoppers often purchase groups of products at the same time. However, a data analyst would need to validate that it makes sense for a recommendation engine to group baby clothes with an order of diapers, applesauce and sippy cups.

Other key differences between supervised and unsupervised learning

- **Goals:** In supervised learning, the goal is to predict outcomes for new data. You know up front the type of results to expect. With an unsupervised learning algorithm, the goal is to get insights from large volumes of new data. The machine learning itself determines what is different or interesting from the dataset.
- **Applications**: Supervised learning models are ideal for spam detection, sentiment analysis, weather forecasting and pricing predictions, among other things. In contrast, unsupervised learning is a great fit for anomaly detection, recommendation engines, customer personas and medical imaging.
- **Complexity:** Supervised learning is a simple method for machine learning, typically calculated through the use of programs like R or Python. In unsupervised learning, you need powerful tools for working with large amounts of unclassified data. Unsupervised learning models are computationally complex because they need a large training set to produce intended outcomes.
- **Drawbacks**: Supervised learning models can be time-consuming to train, and the labels for input and output variables require expertise. Meanwhile, unsupervised learning methods can have wildly inaccurate results unless you have human intervention to validate the output variables.

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Supervised learning vs. unsupervised learning: Which is best for you?

Choosing between supervised and unsupervised learning depends on how your data scientists assess the structure and volume of your data, as well as your situation. To make your decision, be sure to do the following:

- **Define your goals:** Do you have a recurring, well-defined problem to solve? Or will the algorithm need to predict new problems?
- **Review your options for algorithms:** Are there algorithms with the same dimensionality you need (number of features, attributes or characteristics)? Can they support your data volume and structure?

Classifying big data can be a real challenge in supervised learning, but the results are highly accurate and trustworthy. In contrast, unsupervised learning can handle large volumes of data in real time. But, there's a lack of transparency into how data is clustered and a higher risk of inaccurate results. This is where semi-supervised learning comes in.

Semi-supervised learning: The best of both worlds

Can't decide on whether to use supervised or unsupervised learning? **Semi-supervised learning** is a happy medium, where you use a training dataset with both labeled and unlabeled data. It's particularly useful when it's difficult to extract relevant features from data — and when you have a high volume of data.

Semi-supervised learning is ideal for medical images, where a small amount of training data can lead to a significant improvement in accuracy. For example, a radiologist can label a small subset of CT scans for tumors or diseases so the machine can more accurately predict which patients might require more medical attention.

Learn more about supervised and unsupervised learning

Machine learning models are a powerful way to gain the data insights that improve our world. To learn more about the specific algorithms used with supervised and unsupervised learning, we encourage you to delve into the Learn Hub articles on these techniques. We also recommend checking out the blog post that goes a step further, with a detailed look at deep learning and neural networks.

- What is Supervised Learning?
- What is Unsupervised Learning?
- AI vs. Machine Learning vs. Deep Learning vs. Neural Networks: What's the Difference?

To learn more about how to build machine learning models, explore the free tutorials on the IBM Developer Hub.

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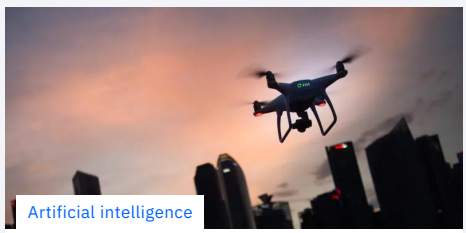

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